

Project Conventions

Routes

- `alerts\_index` → GET `/alerts`
- `clear\_alerts` → POST `/alerts/clear`
- `simulation\_index` → GET `/simulation`
- `buy\_route` → POST `/simulation/buy`
- `sell\_route` → POST `/simulation/sell`
- `backtest\_index` → GET `/backtest`
- `promote\_settings` → POST `/api/promote\_settings`

Templates

- `templates/alerts.html`
- `templates/simulation.html`
- `templates/backtest.html`

Context Variables

- `alerts`: list of dicts `{ id, symbol, price, status, confidence, timestamp, spark, triggers }`
- `backtest\_results`: list of dicts `{ symbol, buy\_time, buy\_price, sell\_time, sell\_price, pl, exit\_reason }`
- `current\_filter`
- `strategy\_params`: dict of tweakable strategy parameters (profit_target, stop_loss, hold_time, min_triggers, etc.)
- `news\_api\_key`

Config Handling

- UI-based parameter controls are available in `/backtest` and (optionally) on the live dashboard.
- Strategy parameters are saved to `config.json` and shared between backtest and live scanner.
- The "Promote to Live" button in the backtest page updates the shared config, instantly applying those parameters to the live scanner.

DB Path

In `services/alert\_service.py`:

```
```python
import os
DB = os.path.abspath(os.path.join(os.path.dirname(__file__), '..', 'alerts.db'))
```
```

- **Backtest results** are *not* stored in the database by default; export as CSV if needed.

Navigation

- All pages feature a nav bar with shortcuts to Trade Alerts, Simulation, Backtest, and Settings (if present).

Full Scanner Simulation

Our full scanner simulation engine iterates through the entire symbol universe, applies all enabled indicator filters (SMA, RSI, MACD histogram, Bollinger Bands breakout, VWAP threshold, etc.), and simulates trades in real-time or historical replay mode. It logs buy/sell events, reasons for decisions, and portfolio performance over time.

Full Scanner Backtesting

The backtesting module replays historical 1-minute and 1-day bar data across the full symbol set, applying the same indicator logic and trade execution rules. It calculates detailed metrics including win rate, average return, maximum

drawdown, and Sharpe ratio—enabling side-by-side comparison of strategy variations.

Complex Indicator Logic

We support a full suite of technical indicators:

- **Simple Moving Average (SMA):** Configurable lookback and cross-over triggers.
- **Relative Strength Index (RSI):** Overbought/oversold thresholds and slope detection.
- **MACD Histogram:** Fast/slow EMA spans and signal line for momentum.
- **Bollinger Bands:** Customizable length and standard-deviation multipliers for breakout detection.
- **Volume Weighted Average Price (VWAP):** Intraday benchmark with threshold offsets.
- **Custom Combinations:** Flexible toggles for mixing any subset of the above filters in any order.